

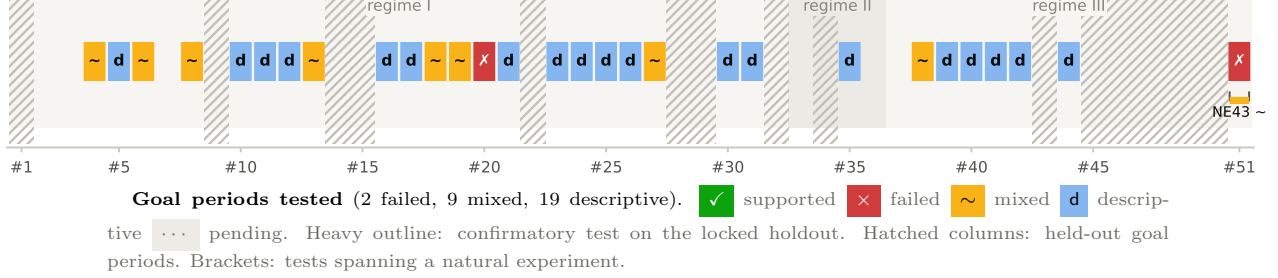
H20: Aging: day-to-day content autocorrelation depends on time since kickoff, not just the lag

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. LLM-swarm content shows no detectable aging at day resolution, in either embedding model: after a few days' kickoff relaxation it decorrelates at an age-independent rate, and only a null calibrated to the low-dimensional fluctuation shape says so honestly.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

We measured the two-time correlation $C(t_w + \tau, t_w)$ of each agent's day-mean content embedding (t_w = active days since kickoff) in 29 non-holdout goal periods. The aging slope A at fixed lag ($t_w \geq 2$) was tested against a fitted stationary swarm model; the pre-registered isotropic null was 2–3× too narrow (content fluctuates in 5–12 of 32 dimensions), so a null with the fitted shape was used. **No aging:** random-effects A over long and medium goals is $+0.021 \pm 0.021$. #38 shows a kickoff relaxation (lag-1 C 0.66→0.87 in four days, then flat); in #51 $\mu = 0.5$ aging is rejected. **Round 1b** (second embedding model, dedupe, style residualization, shared goal vectors): every verdict reproduces in all 7 input configurations; style residualization halves A and removes #51's weak late slowing. Natives: newcomers relax on their own clock without aging; drive withdrawal and a spontaneous rivalry→collaboration turn leave content correlation intact. Holdout not run.

PHYSICAL MODEL

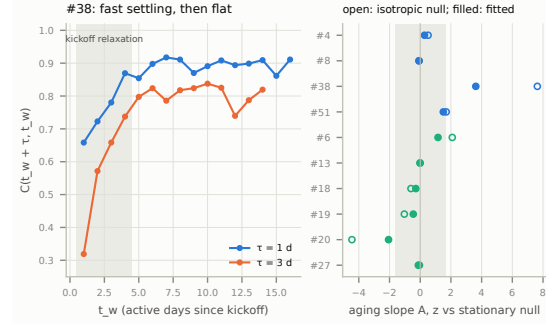
Models 01 (SK-type glassy aging) and 11 (vector spins). Spin: agent i 's day state $\bar{x}_{it}$, the mean of its unit-normalized, regime-whitened ($n = 32$) statement embeddings (own chat and plans); v_{it} is its sampling noise. q : plateau (style and goal content); c_0 : short-lag ceiling; τ_0 : relaxation time; μ : aging exponent of the Box-Cox clock s_μ ($\mu = 0$ stationary, $\mu = 1$ simple aging). Rival: a kickoff transient decaying in τ_q days. n_{eff} : dimension of the day-to-day fluctuations.

$$C(t, t') = \frac{\sum_i \bar{x}_{it} \cdot \bar{x}_{it'}}{\sqrt{\sum_i S_{it} \sum_i S_{it'}}}, \quad S_{it} = |\bar{x}_{it}|^2 - v_{it}$$

$$C(t_w + \tau, t_w) = q + (c_0 - q) e^{-\Delta s / \tau_0}$$

$$\Delta s = s_\mu(t_w + \tau) - s_\mu(t_w), \quad s_\mu(t) = \frac{t^{1-\mu} - 1}{1-\mu}$$

$$A = \partial C / \partial \ln t_w \big|_\tau, \quad \text{Var } C(t, t') \propto 1/n_{\text{eff}}$$



Observables. Left: in #38, the agent-averaged correlation between an agent's content on day t_w and on day $t_w + \tau$ climbs for four active days after the kickoff, then stays flat. A quench relaxes; the system does not slow with age. Right: the aging slope A as a z-score against the stationary null, for long (blue) and medium (aqua) goals. Open circles: the pre-registered isotropic null (too narrow). Filled: the null with the fitted fluctuation shape. Only #38 (up) and #20 (Thanksgiving, down) stand out.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a negative result, mildly useful. An operator worried that a long-running swarm "rigidifies" as its goal ages can relax a little: at day resolution, content turnover does not slow in week three versus week one (weak aging is not excluded). The practical number is the settling time after a big kickoff, about four active days, seen once (#38). The more transferable lesson is methodological: embedding similarity statistics on swarm logs need nulls with the empirical, low-dimensional fluctuation shape, or they are 2–3× overconfident. As physics, aging finds nothing to grip: daily re-prompting with a fixed goal acts like a strong pinning field.

RESULTS VS PREDICTIONS

Prediction	Round 1 $\rightarrow$ 1b (bge / gte)	Verdict
P2 aging in $\geq 2/4$ 1/4 (#38) $\rightarrow$ 1/4 / 1/4 long goals		$\times$
RE A, long + medium	$+0.021 \pm 0.021 \rightarrow +0.020 / +0.022$; style $+0.008$	$\times$
P4 late slowing where $A > 0$	#38 $A_{\text{late}} -0.04 \rightarrow -0.04 / -0.06$	$\times$
#51 $\mu = 0.5$ rejected	power $1.00 \rightarrow \geq 0.98$ in all configs	$\checkmark$
#51 weak $\hat{\mu} > 0$ (descriptive)	$0.22 [0.07, 0.41] \rightarrow 0.16 [0.00, 0.37]$; gone style $CI \ni 0$	
Calibration (robust z-SD)	$1.07 \rightarrow 1.07 / 0.91$	$\checkmark$
G51 newcomers: on-boarding $K > 0$	3 of 5 joiners; no own-clock aging	$\sim$
G27 mid-period switch	$B_{\text{max}} p 0.48 / 0.21$	d
NE43 drives with-drawn	08-05 dip -0.022 inside placebo; 08-21 none	$\sim$

GOAL PERIODS IN DETAIL

G04 $\sim$ 1b: mixed (both models, all configs)
G05 d 1b: descriptive (both models)
G06 $\sim$ 1b: mixed (both models, all configs)
G08 $\sim$ 1b: mixed (both models, all configs)
G10 d 1b: descriptive (both models)
G11 d 1b: descriptive (both models)
G12 d 1b: descriptive (both models)
G13 $\sim$ 1b: mixed (both models, all configs)
G16 d 1b: descriptive (both models)
G17 d 1b: descriptive (both models)
+20 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

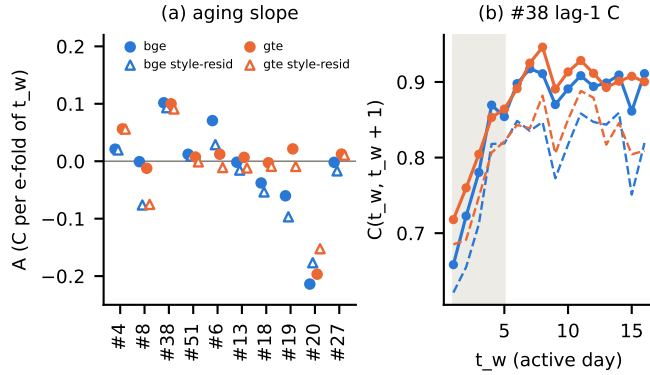
CAVEATS

Day resolution only; weak sub-aging ($\mu \lesssim 0.3$) cannot be excluded outside #51. The calibrated null was chosen after seeing real data, and NE43 shows it is too narrow for step statistics at ordinary day boundaries. Short periods: the null is anti-conservative and holidays dominate. Onboarding transients rest on 5 joiners.

ROUND 2

Sub-day two-time functions on the active-hour clock (link to H48's $\tau \approx 2\text{--}5$ h) and an empirical-placebo null for boundary tests.

- **Essence.** The swarm has no long-term internal memory; all persistence is stored outside the agents.
- **H20-R1** Persistent content order is explained entirely by persistent fields (goal prompt, artifacts); without them content decorrelates within a day (E3, E4).
- **H20-R2** The $\sim 2\text{--}4$ day kickoff relaxation is set by artifact build-up, not agent memory.



Second figure. (a) Aging slope per long and medium goal, bge (blue) vs gte (orange), filled base, open style-residualized: the models agree, style shrinks #51 and #19. (b) #38: lag-1 correlation rises for about four active days (shaded) and then stays flat in both models (dashed: style-residualized).